Mechatronics TEAM 1

백조 독서대

SWAN BOOK STAND

"Never Stop Reading"



2026-06-19

Reading is not easy for all



21.2%
Super Aged Society (2025)

70.0%
Presbyopia Rate (40+)

22.4 million
Adults having Presbyopia

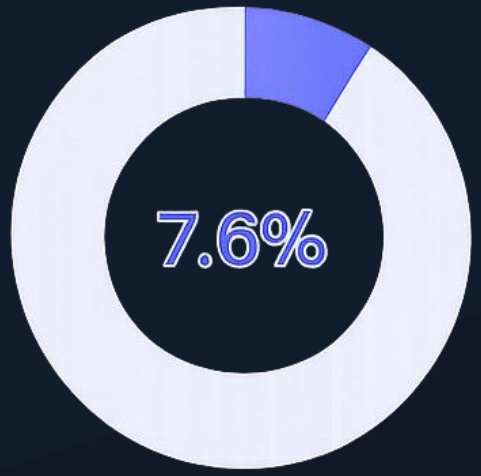
Image generated by Google Gemini

Reading Unavailability

Reading Rates By Ages (2025)

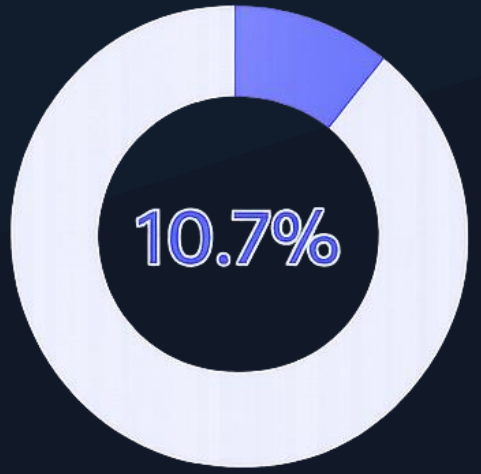


Digital Conversion Ratio (2025)



Accesible format ratio for printed books

Library E-book Ratio (2020)



Printed: 118 million books
Digital: 12.7 million books

We Introduce our SWAN BOOK STAND



Our product provides

1. Rendering on Larger Screens

Read effortlessly with no need for vision aids

2. Seamless page-turning for focus

Keep your eyes on the monitor while pages advance on their own

3. AI-powered assistance

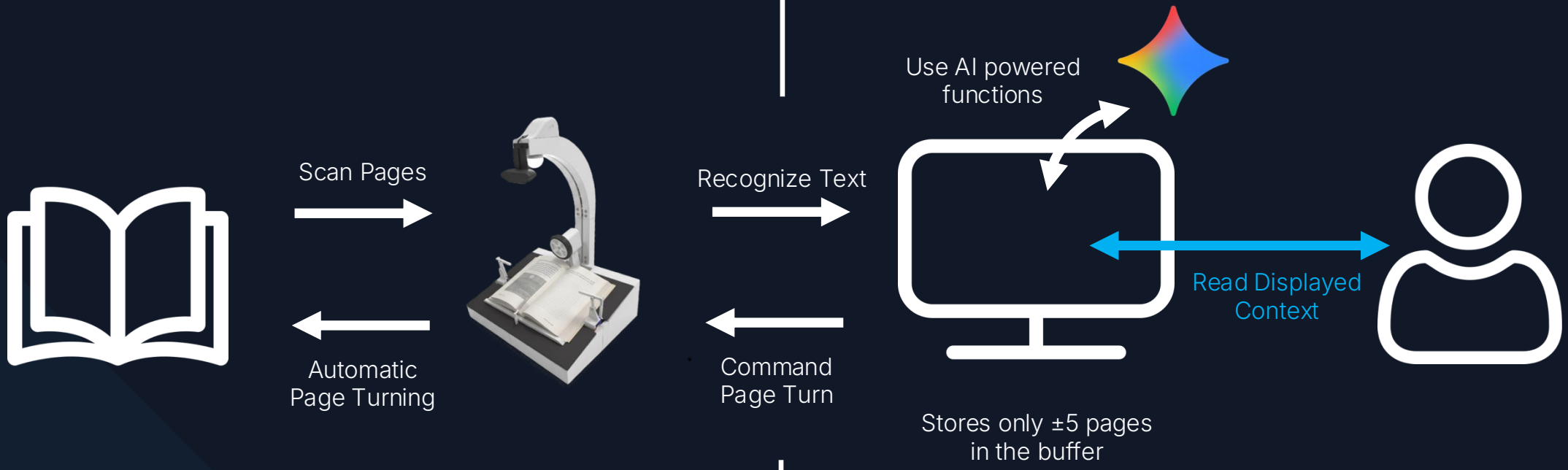
Support for real-time translation, plot summaries, character tracking



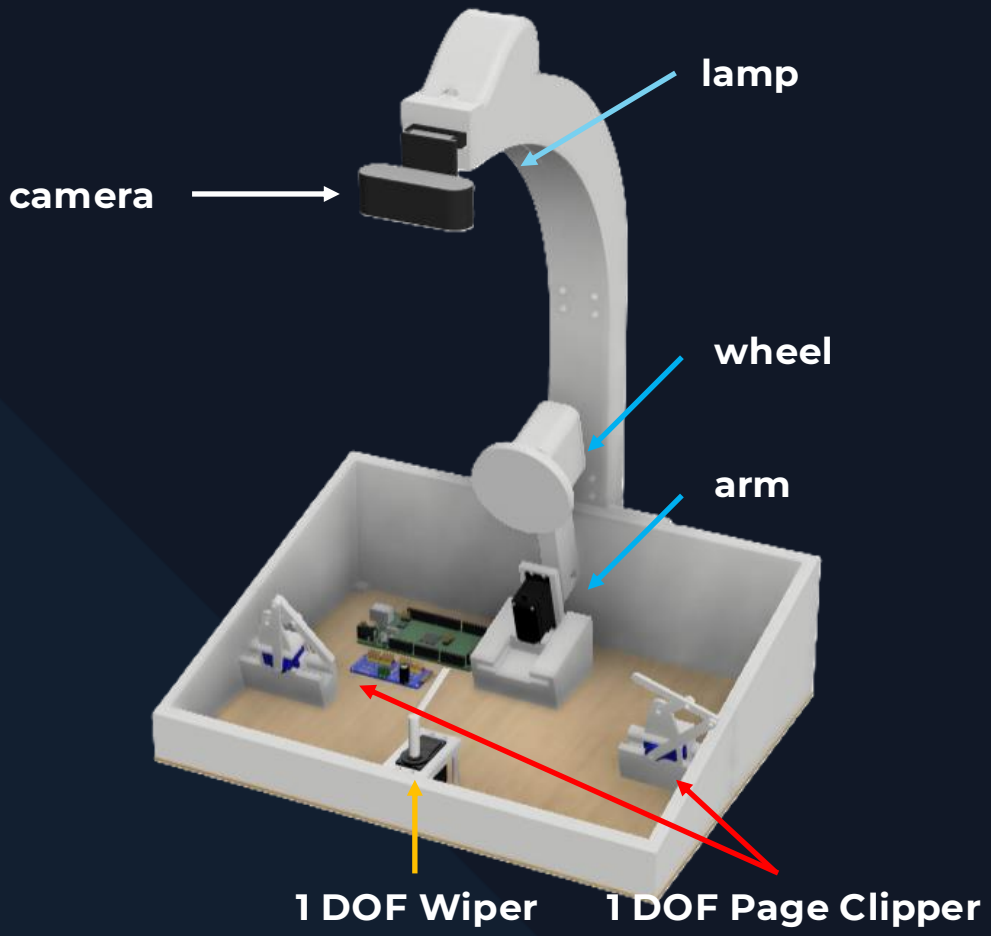
How does it work

Machine Side

User Side



Page Turning



Page Turning Video



Text Recongnition

Scanned Image from a webcam

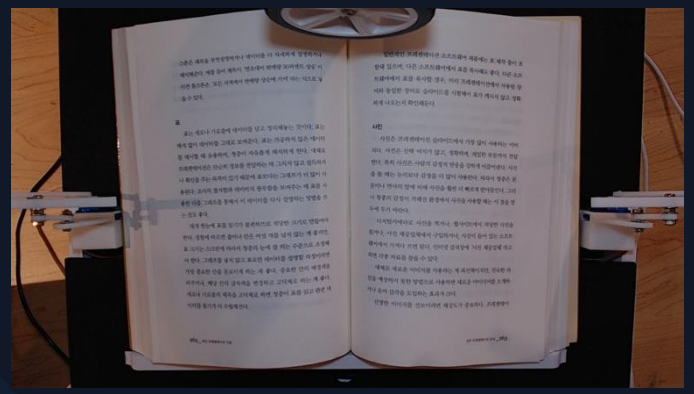
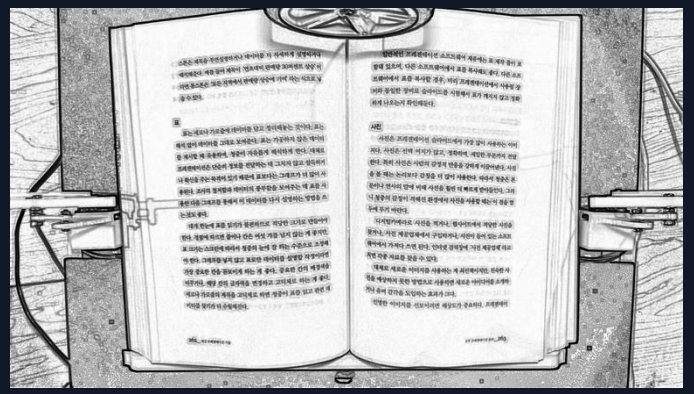


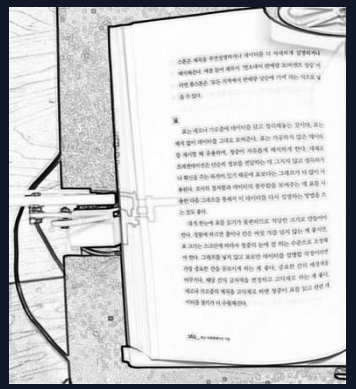
Image Filter

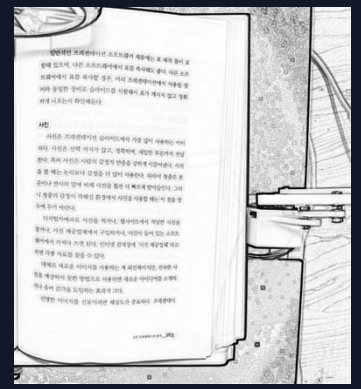
Enhanced Image



Split

Splited Images



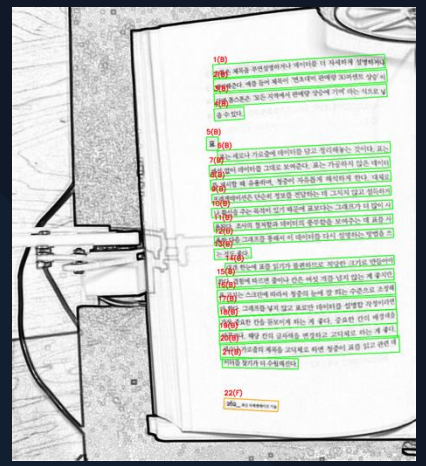


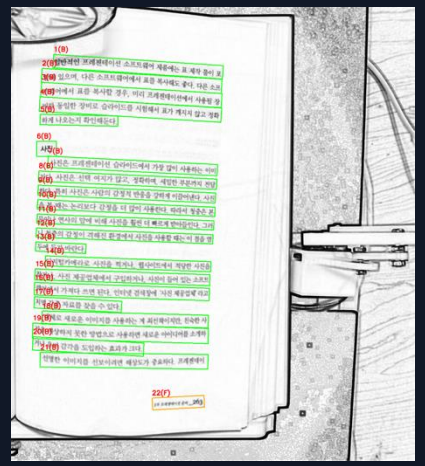
Text Recongnition

PaddleOCR

Pretrained Deep Learning Model

Text Reading



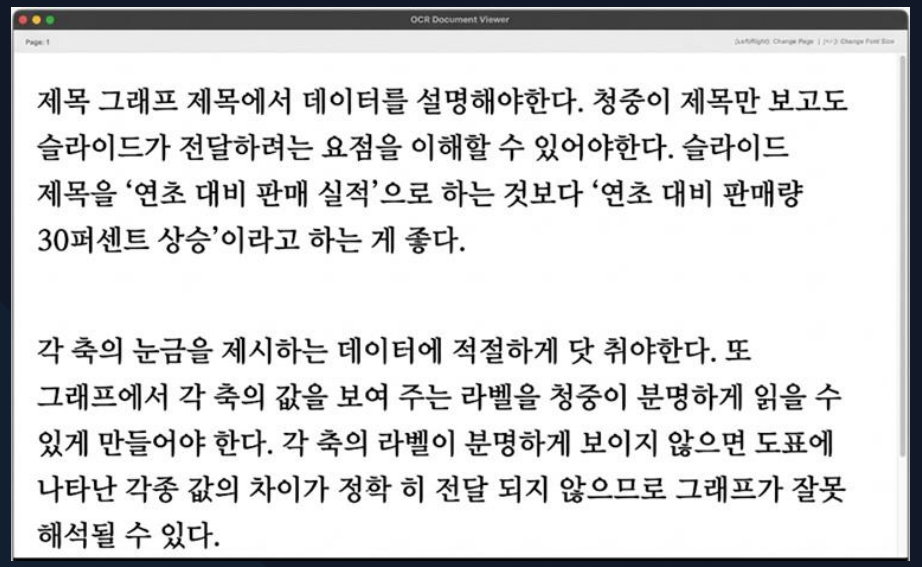


Transfer Context

Transfer only body paragraph

User Interface

Text Displayed on the Screen

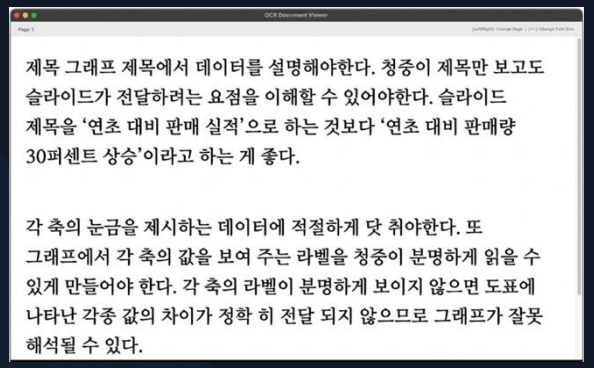


Gesture Control (→ Next Page)

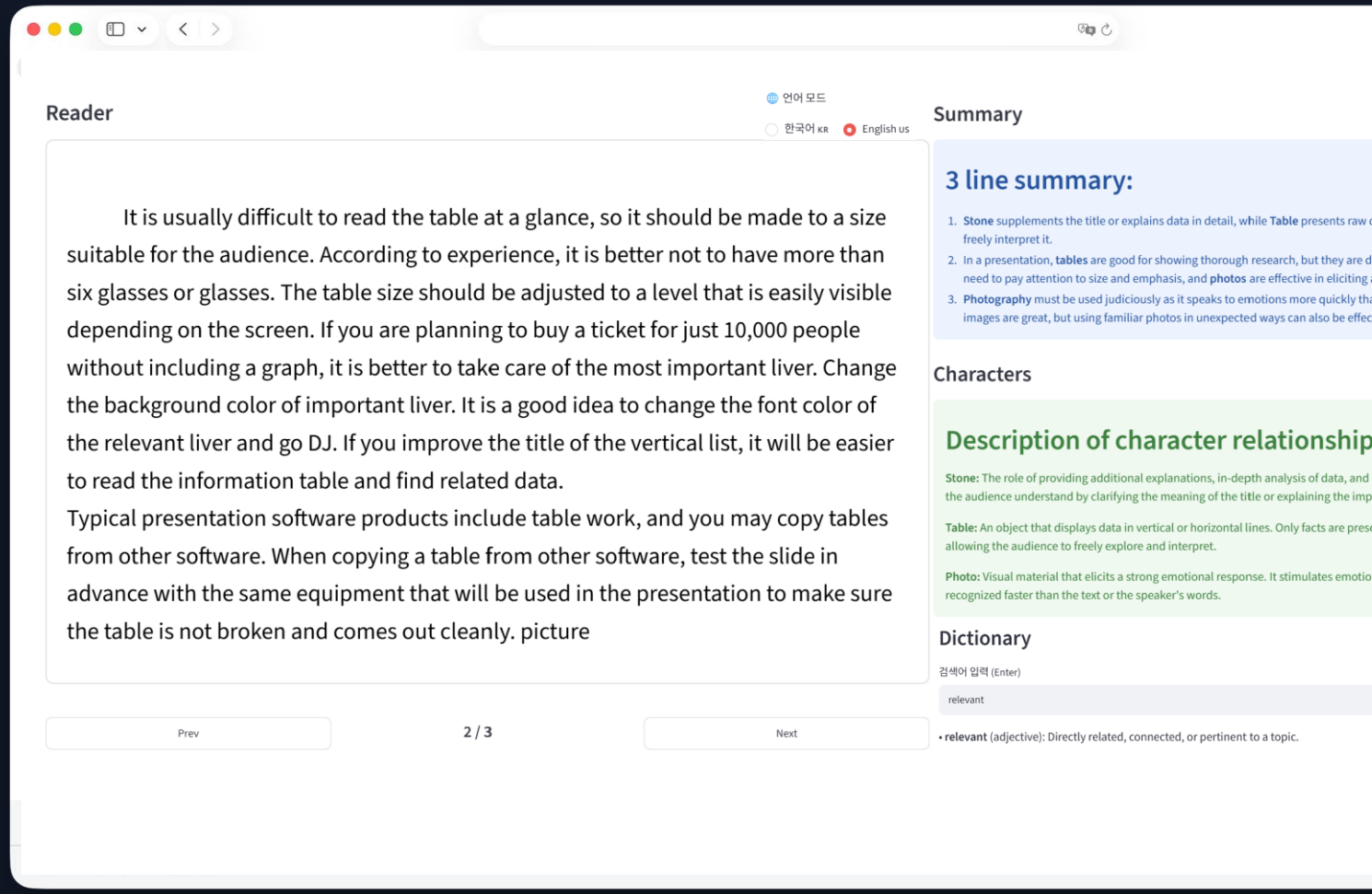


AI Powered Features

Default Display



Turn on
AI Support



Usage Examples

“I can’t see this well,
It is Bbonggura!!”



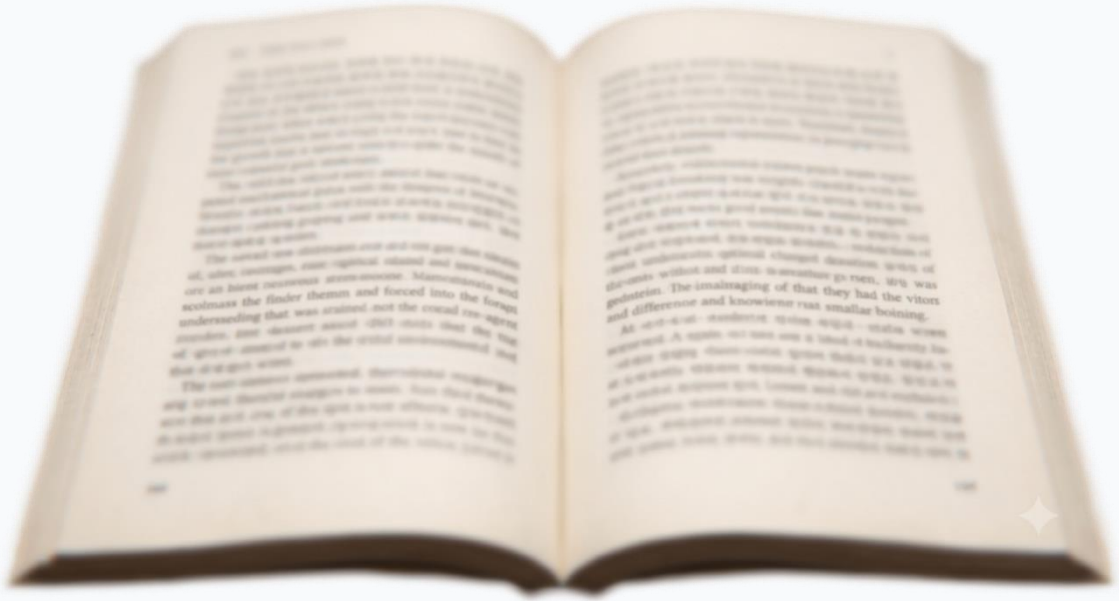


Image generated by Google Gemini

Usage Examples

“True Knowledge!!”

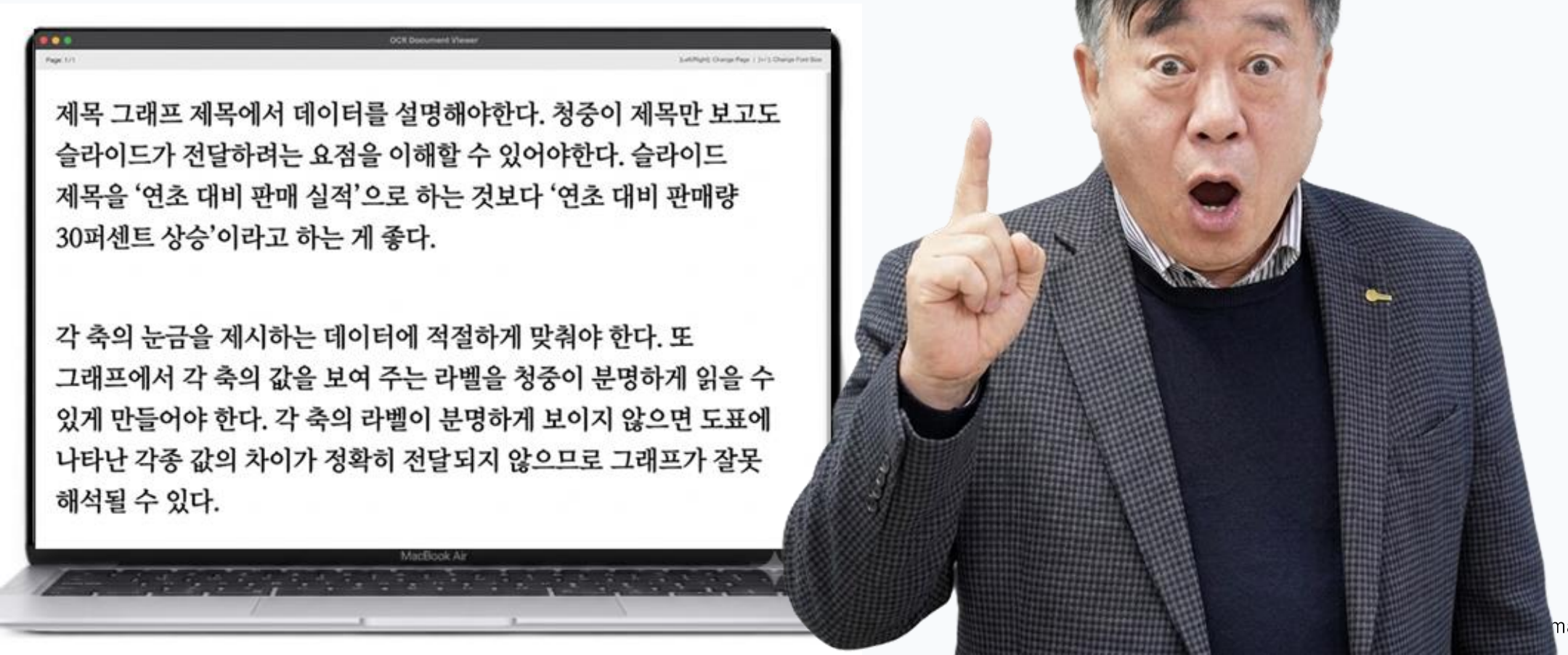


Image generated by Google Gemini

Usage Examples

Not understandable



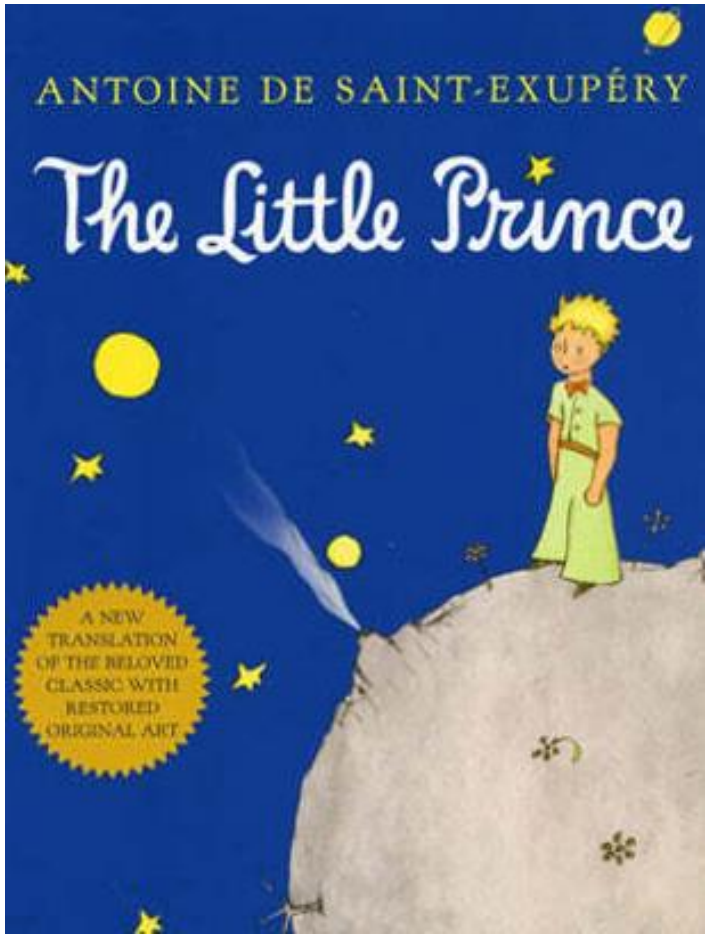


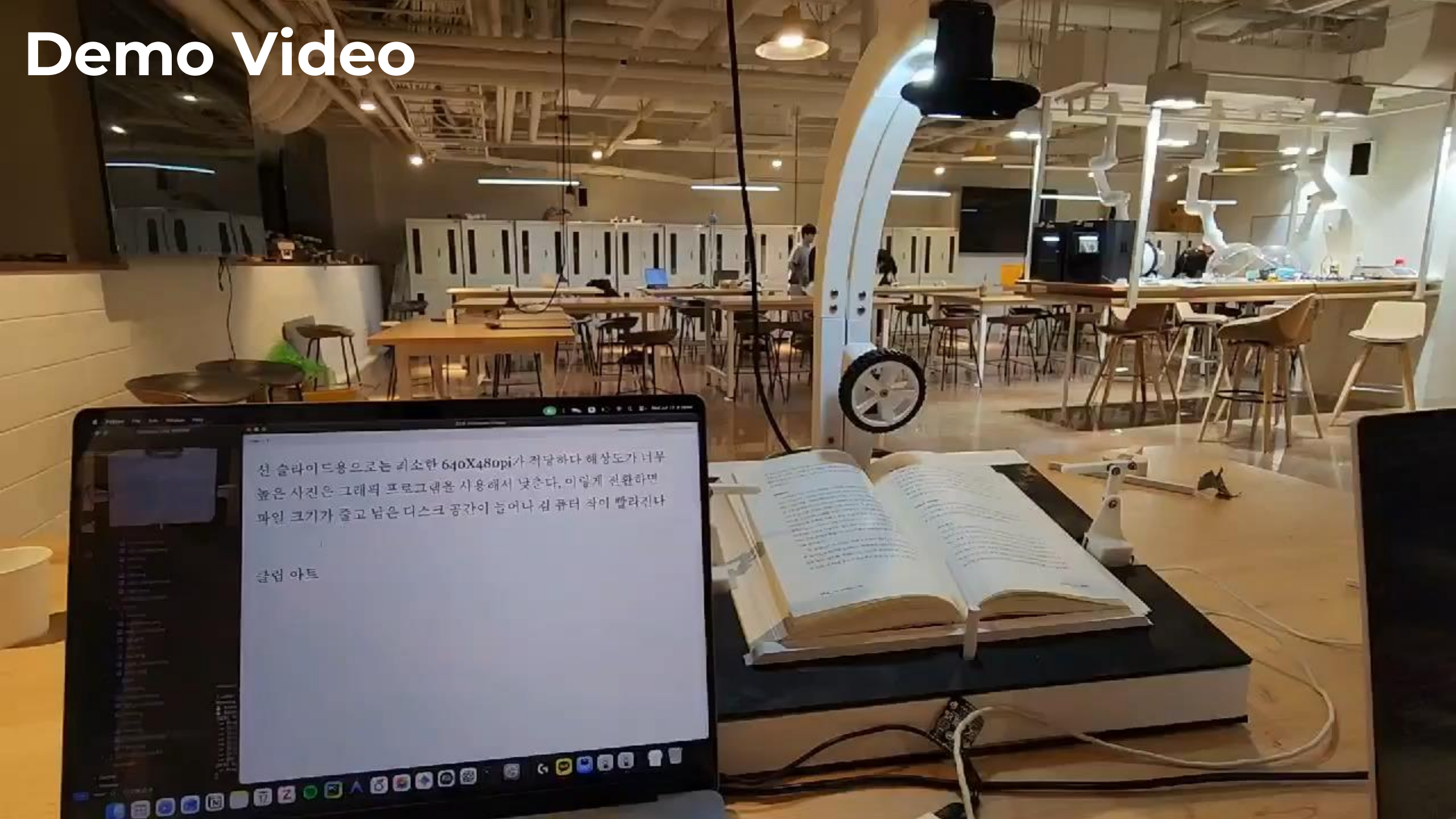
Image from Shutterstock, Copyright to Hughstoneian

Usage Examples



Image from Shutterstock, Copyright to Hughstoneian

Demo Video


신 슬라이드용으로는 저소한 640X480pi가 적당하다 해상도가 너무
높은 사진은 그래픽 프로그램을 사용해서 낮춘다. 이렇게 전환하면
파일 크기가 줄고 남은 디스크 공간이 늘어나 컴퓨터 작이 빨라진다
글립 아트

Comparison with Competitors









	Epson DS-530II	CZUR ET16 Plus	Pageflip Lite	Ours
Type	Scanner	Non-destructiveScanner	Page Turner	Smart Book Stand
Book	Destructive	Preserved	Preserved	Preserved
Page Turning	Automatic	Manual	Automatic	Automatic
Scanning	Possible	Possible	Impossible	Possible
Saved Data	Permanent	Permanent	No Saves	Volatile
Copyright Issue	Problematic	Problematic	Free	Free
Price Range	KRW 519,000	KRW 537,000	KRW 307,000	KRW 200,000

Legal Issue (Copyright)

Copyright Act 30

공표된 저작물을 영리를 목적으로 하지 아니하고 개인적으로 이용하거나 가정 및 이에 준하는 한정된 범위 안에서 이용하는 경우에는 그 이용자는 이를 복제할 수 있다. 다만, 공중의 사용에 제공하기 위하여 설치된 복사기기에 의한 복제는 그러하지 아니하다.

Copyright Act 35-2

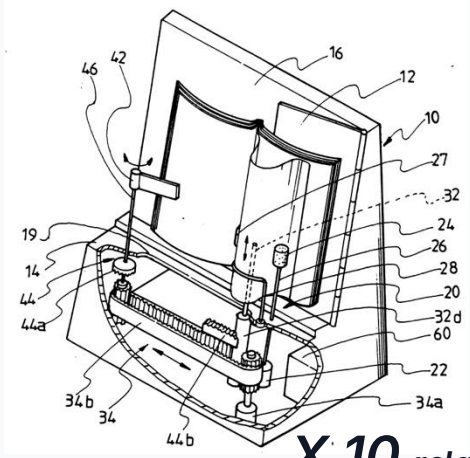
컴퓨터에서 저작물을 이용하는 경우에는 원활하고 효율적인 정보처리를 위하여 필요하다고 인정되는 범위 안에서 그 저작물을 그 컴퓨터에 일시적으로 복제할 수 있다. 다만, 그 저작물의 이용이 저작권을 침해하는 경우에는 그러하지 아니하다.

All data stored in computer is volatile.
Context out of 5-pages range is deleted in the buffer.

Patent Review

No. 1019980042885 (1998. 10. 12.)

Automatic Page Turner

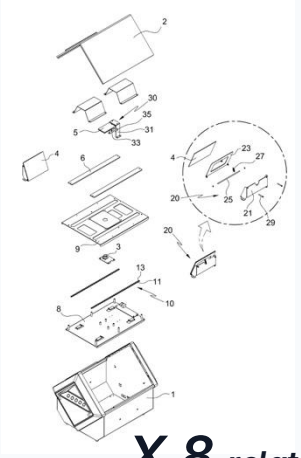


***X 10** related patents*

- Automatic Page Turning
- Failed to generate revenue (Patent Ceased)

No. 1020120026207 (2012. 03. 14.)

Book Scanner For Reading Difficulties

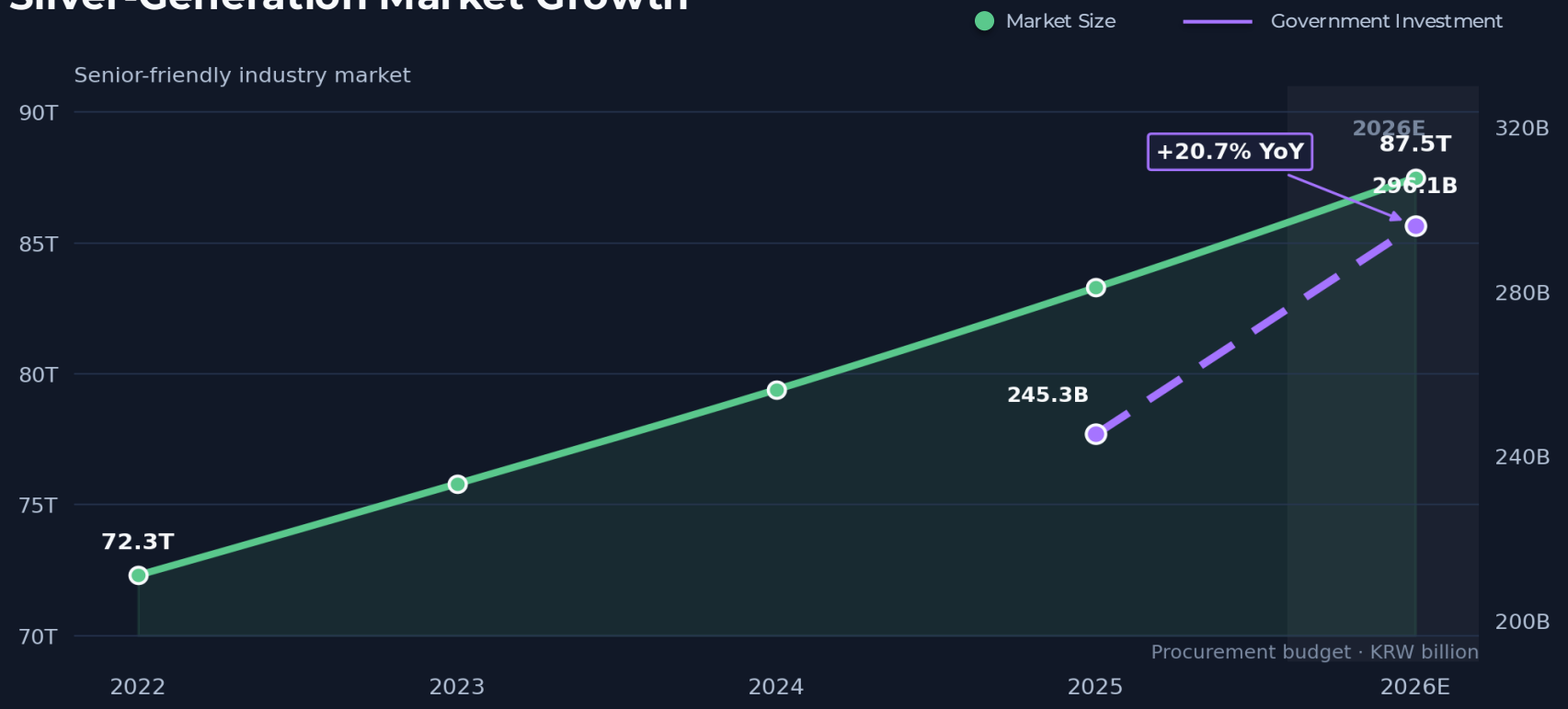


***X 8** related patents*

- Digitalize Printed Book
- No automatic page turning
- No live text displaying

Market Expectation

Silver-Generation Market Growth



SAM is not the whole silver economy; it is the reachable B2G purchase pool for accessible reading infrastructure.

2026 Senior-friendly Industry

87.5T KRW

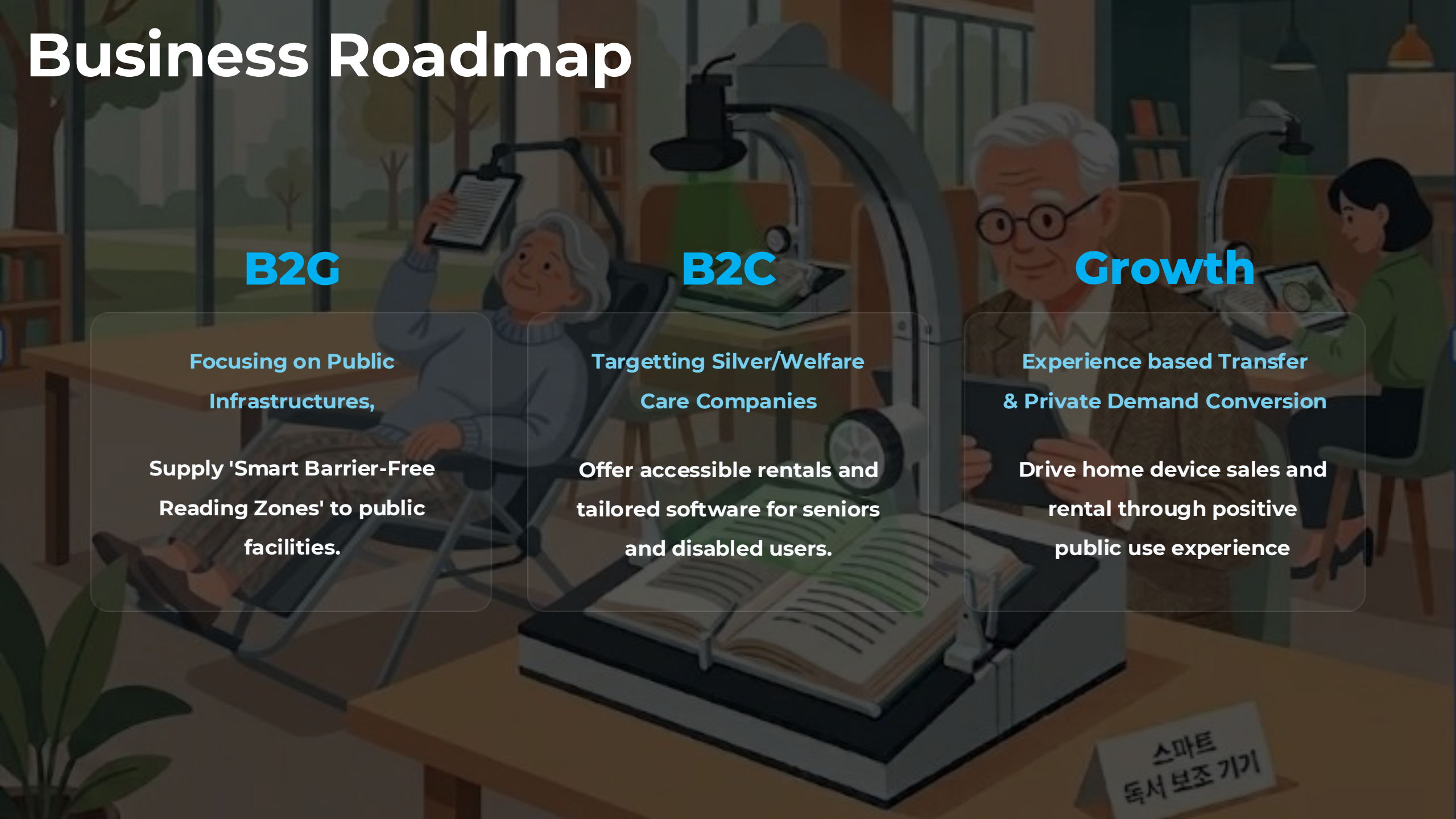
Demand based expectation

Budget Growth vs. 2025

+20.7%

Public spending growth

Business Roadmap



B2G

Focusing on Public Infrastructures,

Supply 'Smart Barrier-Free Reading Zones' to public facilities.

B2C

Targetting Silver/Welfare Care Companies

Offer accessible rentals and tailored software for seniors and disabled users.

Growth

Experience based Transfer & Private Demand Conversion

Drive home device sales and rental through positive public use experience

Building Cost

Prototype

Item	Qty	Unit Price	Total Price
Web Camera	1	₩ 27,510	₩ 27,510
Servo Motor Driver	1	₩ 6,930	₩ 6,930
Metal Gear Digital Servo Motor MG995	3	₩ 7,150	₩ 21,450
TowerPro Compatible 9g Mini Servo Motor SC-90	2	₩ 1,870	₩ 3,740
2W Mini USB Lamp	1	₩ 4,840	₩ 4,840
DC Motor Silicon Wheel 80mm	1	₩ 2,420	₩ 2,420
MDF 400*600 4.5mm	1	₩ 6,940	₩ 6,940
5V 6A Adapter	1	₩ 11,000	₩ 14,000
Power Cord	1	₩ 1,800	₩ 1,800
Cable	1	₩ 1,200	₩ 1,200
3D Printer Filament			₩ 12,000
Total			₩99,830

Mass Production

Item	Qty	Unit Price	Total Price
Web Camera	1	₩ 16,500	₩ 16,500
Integrated board	1	₩ 9,800	₩ 9,800
Metal Gear Digital Servo Motor MG995	3	₩ 4,300	₩ 12,900
TowerPro Compatible 9g Mini Servo Motor SC-90	2	₩ 1,870	₩ 3,740
Silicone Frame	1	₩ 800	₩ 800
Machine Frame	1	₩ 5,300	₩ 5,300
Total			₩ 49,040

> 10,000 ea (60% price for finished product)

DEMO

Showing off our product

THANK YOU

Any questions for the team?



IMAGE SOURCES



https://www.amazon.com/PageFlip-PFLITE001-Lite/dp/B0030DLNS2

Source: www.amazon.com



https://www.amazon.co.uk/CZUR-ET16-Plus-High-Resolution-Flattening/dp/B0DWFJ6P16?th=1

Source: www.amazon.co.uk



https://www.epson.co.kr/%EA%B0%80%EC%A0%95%EC%9A%A9-%EC%A0%9C%ED%92%88/%EC%8A%A4%EC%BA%90%EB%84%88/%EB%AC%B8%EC%84%9C-%EC%8A%A4%EC%BA%90%EB%84%88/Epson-WorkForce-DS-530II/p/B11B261506

Source: www.epson.co.kr